

Lower Serum Testosterone Associated with Elevated Polychlorinated Biphenyl Concentrations in Native American Men

Background Polychlorinated biphenyls (PCBs) and chlorinated pesticides are endocrine disruptors, altering both thyroid and estrogen hormonal systems. Less is known of action on androgenic systems.

Objective We studied the relationship between serum concentrations of testosterone in relation to levels of PCBs and three chlorinated pesticides in an adult Native American (Mohawk) population.

Methods We collected fasting serum samples from 703 adult Mohawks (257 men and 436 women) and analyzed samples for 101 PCB congeners, hexachlorobenzene (HCB), dichlorodiphenyldichloroethylene (DDE), and mirex, as well as testosterone, cholesterol, and triglycerides. The associations between testosterone and tertiles of serum organochlorine levels (both wet weight and lipid adjusted) were assessed using a logistic regression model while controlling for age, body mass index (BMI), and other analytes, with the lowest tertile being considered the referent. Males and females were considered separately.

Results Testosterone concentrations in males were inversely correlated with total PCB concentration, whether using wet-weight or lipid-adjusted values. The odds ratio (OR) of having a testosterone concentration above the median was 0.17 [95% confidence interval (CI), 0.05–0.69] for total wet-weight PCBs (highest vs. lowest tertile) after adjustment for age, BMI, total serum lipids, and three pesticides. The OR for lipid-adjusted total PCB concentration was 0.23 (95% CI, 0.06–0.78) after adjustment for other analytes. Testosterone levels were significantly and inversely related to concentrations of PCBs 74, 99, 153, and 206, but not PCBs 52, 105, 118, 138, 170, 180, 201, or 203. Testosterone concentrations in females are much lower than in males, and not significantly related to serum PCBs. HCB, DDE, and mirex were not associated with testosterone concentration in either men or women.

Conclusions Elevation in serum PCB levels is associated with a lower concentration of serum testosterone in Native American men.